

Reproduction of a research claim from Rinaldi et al. (2016),
from Cognition

Reproduction Team: Samuel Shaki

Research Scientist: Melissa Kline

Independent Reviewer(s)

(add name below when you initiate review, comment "DONE" on your name when you finish
and notify the reproduction team):

Reviewer #1: [NAME]

As necessary:

Reviewer #2: [NAME]

Reviewer #3: [NAME]

Review Period: November 16 - November 23

View-only links to: [Original Paper](#), [Original Materials](#), [Reproduction Materials](#)

Privacy Statement: Other teams are making predictions about the outcomes of many different studies, not knowing which studies have been selected for reproduction. As a consequence, the success of this project requires full confidentiality of this peer review process. This includes privacy about which studies have been selected for reproduction and all aspects of the discussion about these reproduction designs.

Preregistration for SCORE “ADR” Reproductions

Project ID: Rinaldi_Cognition_2016_Kj9d_5196

Study Information

1. Title (provided by SCORE)

RR TEAM INSTRUCTIONS: *This is provided by SCORE.*

Reproduction of a research claim from Rinaldi et al. (2016), from Cognition

2. Authors and affiliations

RR TEAM INSTRUCTIONS: *Fill in the names and affiliations of your team below.*

Samuel Shaki
Ariel University, Ariel, Israel

Description

3. Description of study (provided by SCORE)

RR TEAM INSTRUCTIONS: *This description has been provided by SCORE. If you believe any corrections or clarifications should be made to the claim paragraph or focal test, please let a SCORE team member know.*

The purpose of this reproduction attempt is to assess whether we can reproduce the analytic result as it was reported in the original paper by using the original data. The original data will be analyzed by following the same analytic approach described in the original paper (as close to the same analytic approach as possible based on the level of detail available in the original paper, as well as any supplementary materials or additional information provided by the original author).

All SCORE projects target a single inferential statistical test that supports a statement from the abstract of the original paper. The research claim and focal analysis for this paper are provided below.

Research claim: Visuomotor routines of both finger counting (FC) and reading direction (RD) influence the representation of non-numerical ordered sequences (days of the week), so the fastest mappings result when both routines are compatibly oriented. Reaction times were corrected for finger baseline, and those more than two standard deviations from the overall mean were excluded from the analysis. A 2 x 2 x 2 repeated measures ANOVA with days-direction (thumb-little finger, little finger-thumb), hand posture (prone, supine) and language (English, Hebrew) as within-subjects variables was carried out, yielding a significant triple interaction, $F(1, 19) = 16.69$, $p < .002$, partial eta squared = .47, such that the compatible mappings (Mapping A for English, Mapping G for Hebrew) were the fastest.

Focal inferential test: A significant triple interaction, $F(1, 19) = 16.69$, $p < .002$, partial eta squared = .47, such that the compatible mappings (Mapping A for English, Mapping G for Hebrew) were the fastest.

Summary

Sampling Plan

RR TEAM INSTRUCTIONS: *This section provides a description of the data that will be used for the reproduction.*

4. Existing data (provided by SCORE)

RR TEAM INSTRUCTIONS: *This has been determined by SCORE. Please let a SCORE team member know if this description is not accurate for your reproduction.*

Registration prior to analysis of the data.

5. Explanation of existing data (provided by SCORE)

RR TEAM INSTRUCTIONS: *Please review the information below to make sure it accurately reflects how you will be using the original data. Feel free to add additional details or clarifications about where the data come from or how you will use the data as you see fit.*

This reproduction attempt relies on original (pre-existing) data from the original study. No new data will be created or collected for this reproduction study. The original data were either accessed through an online repository (e.g., OSF, ICPSR, Harvard Dataverse) or another online storage location (e.g., the original author's website or journal website); otherwise the data were provided by the original author(s) directly.

The reproduction team has been provided access to the original dataset, but have only worked with a subset of 5% of the original data in order to write the analysis and complete this preregistration. The reproduction team can reference the 5% dataset subset while writing the analysis to account for any necessary data processing in their code, and they will also use the 5% subset to test their processing and analytic code to confirm that the code runs as expected and produces sensible results (this is further detailed in the Analysis Plan section below). The complete original dataset will not be processed or analyzed until after the reproduction protocol is preregistered.

6. Sample size

RR TEAM INSTRUCTIONS: *Please report below the sample size of the focal analysis selected from the original study, as well as the approximate sample size of the reproduction analysis based on the original dataset provided. Please report as much information here as will be helpful for the review committee to be aware of, including differences in sample size resulting from various analytic decisions (e.g. listwise deletion vs multiple imputation).*

Original study sample size and units: **20 participants**

Reproduction attempt approximate sample size: **20 participants**

Variables

RR TEAM INSTRUCTIONS: *The preregistration form divides variables across three questions: manipulated variables, measured variables, and indices (i.e. analytic variables derived from raw items). Across these questions, you should define all variables that will later be used during your analysis (including data preparation/processing). You can describe all variables in the preregistration and/or summarize and link to a [data dictionary](#) (codebook) on your OSF project.*

*Please **do not** upload the original or reproduction dataset to your OSF project or share it with anyone outside of your reproduction team unless you are sure that doing so would not result in participant identification/data privacy concerns or violation of any copyright restrictions.*

7. Manipulated variables

RR TEAM INSTRUCTIONS: *Manipulated variables in this preregistration refer specifically to variables that have been randomly assigned in an experiment. If your reproduction relies on experimental data, please document each manipulated variable below and use the codebook to indicate what each level of the variable corresponds to (e.g. participants assigned to the treatment condition = 1; participants assigned to the control condition = 0). If your reproduction does not include manipulated variables you may write “NA” below or leave this section blank.*

1. Three manipulated variables in the main task:

(a) Days direction, referred to Sunday–Thursday finger-association (thumb-little finger, little finger-thumb)

(b) Hand posture (prone, supine)

(c) Language (English, Hebrew).

2. Two manipulated variable in the motor baseline task:

(a) The darkened finger

(b) Hand posture (prone, supine)

8. Measured variables

RR TEAM INSTRUCTIONS: *Please use this section to document each variable that was used in the original study’s analysis and the role it served (e.g. dependent variable, control variable, sample parameter, etc). For each variable, provide the description of the variable offered in the paper and/or codebook of the original study and indicate which item(s) in the reproduction dataset it corresponds to. If a reproduction variable is missing from the original dataset, explain how, if at all, you expect its absence will affect the reproduction attempt.*

Specific points to keep in mind (please also consult the [Reviewer Criteria](#)):

- *Does the preregistration document all of the variables needed to reproduce the focal analysis?*
-

1. Response times (RTs) of choosing the correct finger for the day displayed on the screen will be measured for the eight conditions combining three manipulated variables in the main task: (a) Days direction, (b) Hand posture (prone, supine), (c) Language (English, Hebrew).

2. RTs will be measured for two manipulated variables in the motor baseline task: (a) The darkened finger, (b) Hand posture (prone, supine)

9. Indices

RR TEAM INSTRUCTIONS: *If any of the measured variables described in Section 8 will be combined into a composite measure (including simply a mean), describe in detail what measures you will use and how they will be combined. Please be sure this preregistration includes a view-only link to a clearly commented script (uploaded to your OSF project) that constructs the index according to the narrative.*

Specific points to keep in mind (please also consult the [Reviewer Criteria](#)):

- Does the preregistration specify each of the composite measures (e.g. mean scores, factor scores) that are needed for the focal analysis, and which of the measured variables in Section 8 are used in each one (e.g. the happiness, joy, and satisfaction items will be used to create the 'positive feelings' measure)?*
 - Does the preregistration link to a clearly commented script that constructs the indices according to the narrative description?*
-

Note, two irrelevant motor effects can influence RTs: (a) faster motor responses of specific finger over the other fingers, and (b) faster motor responses of prone posture over supine posture. As in the original study, and to avoid any possible confounding motor and hand posture effects, I will:

(a) Calculate for each participant the average RT of each finger in each posture in the appropriate (prone or supine) motor baseline condition. Then, the RT of the fastest finger will be subtracted from the RT of each other finger, obtaining an index of each finger relative speed. This finger index will then be subtracted from the average RT of each finger in each experimental mapping, resulting in a corrected finger RT.

(b) Although the above finger index was done separately in each mapping, the built-in advantage of prone over supine position is still not treated. Hence, for each participant the average RT of each finger in the prone posture will be subtracted from its average RT in the supine posture of the motor baseline condition. These RT will then be subtracted

from the average RT of each finger in each experimental mapping of the supine posture, resulting in a corrected supine finger RT. Importantly, incorrect responses are defined when the finger associated to the stimulus (e.g., day) presented - with reference to the specific mapping tested - is not pressed (hence, any other finger is pressed).

Analysis Plan

10. Statistical models

RR TEAM INSTRUCTIONS: *This section should describe in detail the analysis that will be performed to reproduce the focal result. This analysis must align as closely as possible with the original study's analysis, even if you have identified limitations in the original study. The level of detail should allow anyone to reproduce your analyses from your description below. Examples of what should be specified: the model; each variable; adjustments made to the standard errors and to case weighting; additional analyses that are required to set up the focal analysis; and the software used.*

Please provide a clearly commented script that performs the analysis described in the narrative provided below. The script/syntax should be written in the same language/use the same program(s) as the original study (if possible). Please upload all data processing and analysis scripts/syntax to the "Analysis" component in your OSF project and provide a view-only link to this component (or to specific files) below.

Please also test that the code runs without error on a random subset of 5% of the replication dataset, and provide verification that the code has produced a sensible result by providing a screenshot of the output below (please upload the screenshot and/or output file to OSF as well). Finally, please confirm that you have only developed and tested your analysis plan and code using 5% of the data.

Beyond the reproduction of the focal analysis from the original study, it is at your discretion to test the claim using other analytic approaches as a check of the robustness of the claim. The original analysis should be detailed first and be clearly distinguished from any other analyses.

Specific points to keep in mind (please also consult the [Reviewer Criteria](#)):

- *Does the preregistration specify which statistical model will be used to provide the 'focal evidence' for the SCORE test (e.g. a regression coefficient in a larger multiple regression model), and does it correspond closely to the model and evidence from the original study?*
- *Does the preregistration describe each variable that will be included in the focal analysis, and what role each variable has (e.g. dependent variable, independent variable)?*

- Does the preregistration include a detailed specification of the focal analysis, including interactions, lagged terms, controls, etc., in both narrative form and in a clearly commented script?
- Does the preregistration verify that the code runs without error on a random subset of the replication dataset?

(a) Model description, analysis code, and test verification

Analysis description:

The study uses a 3 factor within-subjects design which manipulates days-direction (thumb-little finger, little finger-thumb), hand posture (prone, supine) and language (English, Hebrew).

A 2 x 2 x 2 repeated measures ANOVA with days direction (thumb-little finger, little finger-thumb), hand posture (prone, supine) and language (English, Hebrew) as within-subjects variables will be carried out. The dependent variable is the corrected RTs (as was explained above, see indices section).

View-only link to data processing and analysis scripts/syntax:

https://osf.io/y5rs2/?view_only=82cdcf282b84e8f8464dd222d6c40df

	Huynh-Feldt	2662.812	4.000	665.703			
	Lower-bound	2662.812	4.000	665.703			
Language * Posture * Direction	Sphericity Assumed	5186.247	1	5186.247	10.255	.033	.719
	Greenhouse-Geisser	5186.247	1.000	5186.247	10.255	.033	.719
	Huynh-Feldt	5186.247	1.000	5186.247	10.255	.033	.719
	Lower-bound	5186.247	1.000	5186.247	10.255	.033	.719
Error(Language*Posture*Direction)	Sphericity Assumed	2022.883	4	505.721			
	Greenhouse-Geisser	2022.883	4.000	505.721			
	Huynh-Feldt	2022.883	4.000	505.721			
	Lower-bound	2022.883	4.000	505.721			

Test verification:

This statement confirms that only 5% of the data have been randomly sampled in developing the analysis plan and code contained in this preregistration.

(b) Known Deviations

RR TEAM INSTRUCTIONS: *Please use this section to document known deviations between the analysis used to test the selected claim from the original study and the analysis documented above. Please explain why these deviations are necessary and what effect they may have on the reproduction outcome.*

There are no known deviations from the original study.

11. Transformations

RR TEAM INSTRUCTIONS: *This section should describe how any of the measured variables or composite measures mentioned above will be transformed prior to the analyses detailed in Section 10. These are any adjustments made to variables in the original dataset, and might include centering, logging, lagging, rescaling, etc. Please provide enough detail such that anyone else could reproduce the transformations based on the description below. Please be sure this preregistration includes a view-only link to a clearly commented script that performs the transformations described in the narrative provided below.*

Specific points to keep in mind (please also consult the [Reviewer Criteria](#)):

- *Does the preregistration specify which of the original variables will need to be transformed prior to the focal analysis?*
 - *For each variable needing transformation, does the preregistration adequately describe the transformations, including any centering, logging, lagging, recoding, or implementation of a coding scheme for categorical variables?*
 - *Does the preregistration link to a clearly commented script that performs each transformation?*
-

As in the original study, and to avoid any possible confounding motor and hand posture effects, I will:

(a) Calculate for each participant the average RT of each finger in each posture in the appropriate (prone or supine) motor baseline condition. Then, the RT of the fastest finger will be subtracted from the RT of each other finger, obtaining an index of each finger relative speed. This finger index will then be subtracted from the average RT of each finger in each experimental mapping, resulting in a corrected finger RT.

(b) Although the above finger index was done separately in each mapping, the built-in advantage of prone over supine position is still not treated. Hence, for each participant

*the average RT of each finger in the prone posture will be subtracted from its average RT in the supine posture **of the motor baseline condition**. These RT will then be subtracted from the average RT of each finger in each experimental mapping of the supine posture, resulting in a corrected supine finger RT. Importantly, incorrect responses are defined when the finger associated to the stimulus (e.g., day) presented - with reference to the specific mapping tested - is not pressed (hence, any other finger is pressed).*

12. Inference criteria (provided by SCORE)

RR TEAM INSTRUCTIONS: *This section describes the criteria that will be used to assess whether the original focal test results listed above were reproduced by the analyses described in Section 10. The default language below only applies to the test of the SCORE claim. It is at your discretion to describe the inferential criteria you will use for any additional analyses.*

Inference criteria for a reproduction will be based on a subset of the following criteria:

- Sample size
- Focal variable p value
- Control variable(s) coefficient(s) in focal model
- Focal variable coefficient value
- Effect size of focal variable
- Subjective interpretation by analyst

Accordingly, the reproduction analysis will report outcomes for each of these criteria whenever possible and relevant.

13. Data exclusion

RR TEAM INSTRUCTIONS: *The section below should describe the rules you will follow to exclude observations from the analyses described in Section 10. Note that this only refers to any exclusions you will make to the original dataset; not any exclusions detailed in the original paper that were already made to the original dataset before it was accessed. Please be as detailed as possible in describing the rules you will follow (e.g., What is the specific definition of outliers you will use? Exactly how many attention checks does a participant need to fail before their removal from the analytic sample?).*

Specific points to keep in mind (please also consult the [Reviewer Criteria](#)):

- *Does the preregistration comment on whether any observations included in the original dataset will be excluded prior to data analysis?*

- *If yes, does the preregistration provided detailed instructions on how the exclusions will be performed (e.g. Is the definition of outlier provided? Is the number of attention checks failed before a participant is excluded specified?)*
-

RTs more than two standard deviations from the overall mean will be excluded from the analysis.

14. Missing data

RR TEAM INSTRUCTIONS: *The section below should describe how missing or incomplete data will be handled. Please be as detailed as possible in describing the exact procedures you will follow (e.g., last value carried forward; mean imputation) and any software required (e.g., “we will use Amelia II in R to perform the imputation”).*

Specific points to keep in mind (please also consult the [Reviewer Criteria](#)):

- *Does the preregistration comment on how missing or incomplete data will be addressed (e.g. casewise removal, missing data imputation)?*
 - *If applicable, does the preregistration specify how many missing variables will lead to a case’s removal (e.g. If a subject does not complete any of the three indices of tastiness, that subject will not be included in the analysis.)?*
 - *If applicable, does the preregistration describe how missing data imputation will be performed, including relevant software?*
-

I’ll use the original raw data in this replication, so no missing data should be in this analysis.

15. Exploratory analysis (Optional)

RR TEAM INSTRUCTIONS: *If you plan to explore your dataset to look for unexpected differences or relationships, you may describe those tests here. An exploratory test is any test where a prediction is not made up front, or there are multiple possible tests that you are going to use. A statistically significant finding in an exploratory test is a great way to form a new confirmatory hypothesis, which could be registered at a later time.*

Additionally, if after running your pre-registered plan, you are unable to produce the focal test, or if the focal test value is substantially different from the reported value, please describe your plan for conducting exploratory analyses that may produce the focal test or originally reported value.

NA

16. Other

RR TEAM INSTRUCTIONS: *This section serves two purposes. First, please use this section to discuss any features of your reproduction plan that are not discussed elsewhere. Literature cited, disclosures of any related work such as replications or work that uses the same data, plans to make your data and materials public, or other context that will be helpful for future readers would be appropriate here. Second, please also re-surface any major deviations from earlier in the preregistration that you expect a reasonable reviewer could flag for concern. Give a summary of these deviations, focusing on larger changes and any possible challenges for comparing the results of the original and the reproduction study.*

NA

Final review checklist

REVIEWER INSTRUCTIONS: *Reviewers -- for the following questions, please indicate whether you can 'sign off' on the following items by adding a comment. You can update this response as the lab moves through revisions during the review period!*

- Included with this preregistration is a narrative description of how the reproduction dataset will be used to perform the focal reproduction analysis, as well as the specific analytic scripts/code/syntax that will be used:
 - Is the analysis plan (including code) that's documented in the preregistration consistent with a high-quality, good-faith reproduction of the focal claim selected from the original study? Did the provided view-only link (VOL) direct to analysis code/syntax, and did that look sufficient for the pre-registered analysis plan?
 - Has the data analyst demonstrated that the analysis code works as expected on a random 5% of the final reproduction dataset?
- I have reviewed all sections of this preregistration, and I believe it represents a good-faith reproduction attempt of the original focal claim.